



ISP Environmental Solutions Pvt. Ltd.
RESTRICTED MANUFACTURER DOCUMENTATION

Manufacturer Technician Guide

Full engineering hotkey, diagnostics, serial safety, recovery and deployment reference for RVMDesktopApp and PecoDropDesktopApp

ISP-AUTHORIZED MANUFACTURER TECHNICIANS ONLY. Direct servo and system-power controls must not be distributed to public users.

Products: Public RVM and Enterprise PecoDrop

Platform: .NET 8 WPF / Windows kiosk

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1. Product boundary and safety

Product	Path	Identity
Public RVM	RVMDesktopApp	Plastic, cans, UBC and rejects; public rewards and leaderboards.
Enterprise PecoDrop	PecoDropDesktopApp	Dual landscape displays, QR/keypad,  plastic,  metal/cans and  weighed paper.

Never mix: machine IDs, serial ports, configuration, ads, organization bindings, credentials, logs or databases. Isolate public access and moving mechanisms before service.

- Use lockout/tagout for mains, motors and cabinet work.
- Keep hands, tools and items clear before any gate command.
- Only one process may own the serial port; AdminWindow reuses the kiosk connection.
- Log command, result, machine ID and final safe state.

2. Common hotkeys

Numeric codes use the focused kiosk window’s digit buffer. Top-row and numeric-keypad digits are accepted. Repeated-key actions should be entered promptly.

Code/key	Product	Function	Verification
0	Both	Start/wake normal session.	Ready state and correct availability.
Enter	Both	Finish session / reward flow.	Correct totals and user context.
111	Both	Open Admin login.	PIN remains required.
888	Both	Toggle telemetry panel.	Observe serial/API/DB messages.
33	Both	Manual chamber-clear command.	Area isolated; verify ready state.
1122	Both	Enable Demo Mode and simulator.	Out-of-service/test status.
1218	Both	Restart confirmation dialog.	Finish sessions before Yes.
1219	Both	Shutdown confirmation dialog.	Use only for intended shutdown.

3. Development keyboard controls

Key	Product	Action
B	RVM	Simulate accepted medium plastic.
C	RVM	Simulate accepted medium can.
U	RVM	Simulate accepted medium UBC.
R	RVM	Simulate rejection.
S	Both	Stop machine session.
A	Both	Open Admin login.
F2 or D	PecoDrop	Open demo-testing window.
Esc	Both	Close telemetry first; otherwise exit application. Landscape PecoDrop closes related windows/services.

Development keys are bench/demo controls, not public or production operating instructions.

4. PecoDrop machine-control codes

PecoDrop only. The current public RVM does not implement this same numeric machine-control set.

Code	Action	Required post-check
66	START machine	Connected, calibrated, compartments available.
77	STOP machine	Mechanisms close; state stopped.
88	RESET hardware	Handshake returns; recalibrate.
99	CALIBRATE	Wait for CALIBRATION:OK .

5. PecoDrop direct servo controls

Moving-mechanism hazard: isolate the kiosk, stop intake and keep the mechanism empty. Follow each OPEN test with the matching CLOSE. Closing Admin or host-lease loss closes all servos.

Code	Mechanism	Serial command
601	Plastic iris open	SERVO:PLASTIC:IRIS:OPEN
602	Plastic iris close	SERVO:PLASTIC:IRIS:CLOSE
603	Plastic drop open	SERVO:PLASTIC:DROP:OPEN
604	Plastic drop close	SERVO:PLASTIC:DROP:CLOSE
605	Metal iris open	SERVO:METAL:IRIS:OPEN
606	Metal iris close	SERVO:METAL:IRIS:CLOSE
607	Metal drop open	SERVO:METAL:DROP:OPEN
608	Metal drop close	SERVO:METAL:DROP:CLOSE
609	Paper iris open	SERVO:PAPER:IRIS:OPEN
610	Paper iris close	SERVO:PAPER:IRIS:CLOSE
611	Paper drop open	SERVO:PAPER:DROP:OPEN
612	Paper drop close	SERVO:PAPER:DROP:CLOSE

1. Close every opened mechanism and visually confirm position.
2. Exit manual testing; verify normal intake remains stopped.
3. Reset/calibrate as required and verify all compartment states.

6. PecoDrop display layout

Code	Result
0012	Hardware kiosk LEFT; video signage RIGHT.
0021	Video signage LEFT; hardware kiosk RIGHT.

Preference is machine-scoped in `dbo.KioskDisplaySettings`; `config.txt` is the offline fallback. Verify fullscreen windows, touch mapping, reboot persistence and HDMI reconnect.

7. Arduino host safety and calibration

```
Firmware boot → HOST:WAITING
Desktop connects → renew HOST:ALIVE
Desktop authorizes → HOST:START
Calibration request/retry → CALIBRATION:OK
Verified compartment query → intake enabled
No host lease for 5 seconds → safe closed state
```

- Firmware: `PecoDropDesktopApp/Arduino/RVM_Arduino/RVM_Arduino.ino`.
- Target Arduino Mega 2560 / ATmega2560.
- Close PecoDrop before firmware upload.
- After hot-plug/relaunch, reconnect, calibrate and verify state before intake.
- `stk500v2_getsync()` is a port/bootloader/upload-path problem, not sketch-size failure.

8. Diagnostic matrix

Symptom	Checks	Recovery
Hardware offline	Power, USB, Device Manager, baud/port, serial owner.	Reconnect; telemetry; reset then calibrate.
False full-bin	Obstruction, startup settling, wiring, calibration.	Clean/empty; reset; calibrate; verify post-calibration.
Paper weight unstable	HX711/load-cell power, wiring, preload, vibration.	Empty, stabilize, calibrate and verify known weight.
Gate incomplete	Binding, supply voltage, linkage, command response.	Power-isolate mechanical work; test one servo at a time.
API/DB offline	Network, DNS/TLS/time, authorization, SQL config.	Preserve offline queue; restore; verify no duplication.

9. Build, data and deployment

```
dotnet publish .\RVMDesktopApp\RVMDesktopApp.csproj -c Release -r win-x64 --self-contained true -
p:PublishSingleFile=false
dotnet publish .\PecoDropDesktopApp\PecoDropDesktopApp.csproj -c Release -r win-x64 --self-contained true -
p:PublishSingleFile=false
```

- Never compile Debug builds on production kiosks.
- Use fixed signed paths, separate ProgramData folders and bounded watchdog recovery.
- PostgreSQL `rvmpg` is the production source of truth.
- MongoDB is read-only/sync-only: never add, edit, update or delete there.
- Never put passwords or tokens in documentation.

10. Return-to-service checklist

- ☐ Cabinet closed; tools removed.
- ☐ All gates/irises physically closed.
- ☐ Correct app and machine ID.
- ☐ Demo Mode closed.
- ☐ Single stable serial connection.

- ☐ Calibration successful.
- ☐ Compartment states verified.
- ☐ Displays/touch mapping correct.
- ☐ API/DB and offline queue checked.
- ☐ Controlled acceptance/rejection test passed.
- ☐ Reward record verified without duplication.
- ☐ Service log completed.